GovShield Sentinel Grid 3.0

Comprehensive Technical Specification & Stack Dossier

Real-Time Multi-Signal AI/ML Cyber Defense & Deceptive Government Portal Neutralization

SIH PROBLEM: SIH1454

RELEASE V3.2.0 (PRODUCTION)

Date: September 2026 | CERT-In Standard

1. Executive Summary & Problem Scope

GovShield Sentinel Grid 3.0 is an enterprise sovereign cybersecurity ecosystem engineered to neutralize zero-day typosquats, credential harvesting phishing operations, and deceptive government clone portals that impersonate Government of India public administration infrastructure (e.g., `.gov.in` and `.nic.in`). Developed for **Smart India Hackathon Problem Statement SIH1454**, GovShield bridges browser endpoint interception, UX4G 3.0 citizen accessibility, a 15-stage defense-in-depth forensic pipeline, explainable soft-voting machine learning ensembles, and cryptographic Proof-of-Authority (PoA) blockchain evidence anchoring certified under **Section 65B of the Indian Evidence Act**.

2. End-to-End System Architecture

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|                                       GOVSHIELD SENTINEL GRID 3.0 ARCHITECTURE                          |
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| [ENDPOINT INTERCEPTION]                                                                           |
|   Chrome MV3 Extension (Active Tab Listener) <---> In-Page Tricolor Fraud Banner / Safe Toast     |
|   UX4G 3.0 Web Portal (12 Indic Languages + WCAG AAA Accessibility Controls + Voice Synthesis)    |
|                                           | (JSON REST / WebSocket)                               |
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| [EDGE ROUTING & ASYNC ORCHESTRATION]                                                              |
|   FastAPI 0.115+ (ASGI Engine) <---> Memory-Bounded Eviction Cache (LRU) & ThreadPoolExecutor     |
|   Dual-Tier Routing: Local Dev Probe (1000ms) --> Cloud Render Failover (8000ms) --> Client Heuristic |
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| [15-STAGE MULTI-SIGNAL FORENSIC PIPELINE]                                                         |
|   1. Canonical URL Normalizer          6. DOM Sensitive Form Harvester      11. Threat Intelligence Hub    |
|   2. Homoglyph & Zero-Width Detector   7. Visual Perceptual Hash (pHash)    12. Content Jaccard Similarity |
|   3. Levenshtein Typosquat Engine      8. HTTP/JS Redirect Unroller         13. OSINT Internet Search Engine |
|   4. Sovereign Brand Matcher (25+)     9. DNSSEC / MX Mail Security         14. Proof-of-Authority Blockchain |
|   5. SSRF-Shielded Safe Crawler        10. RDAP / NIC Sovereign Domain Age  15. Explainable Sovereign ML |
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| [EVALUATION & PROOF GENERATION]                                                                   |
|   Sovereign ML Ensemble (XGBoost 250 trees + Random Forest 180 trees) -> Probability Score        |
|   Bayesian Fusion Engine -> Calibrated Threat Score (0-100) & Verdict (LEGITIMATE / SUSPICIOUS / PHISHING) |
|   Cryptographic Blockchain Anchoring (RFC 8785 Canonical JSON) -> Section 65B CERT-In Incident Dossier |
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3. High-Level Technology Stack Matrix

Subsystem	Primary Technologies & Frameworks	Key Modules / Libraries	Compliance / Standards
Backend API Core	Python 3.11, FastAPI 0.115, Uvicorn, asyncio	Pydantic v2, Starlette, Requests, ThreadPoolExecutor	REST JSON RFC 8785
AI / ML Ensemble	Scikit-Learn, XGBoost, Random Forest	Joblib, NumPy, Pandas, Google GenAI SDK (Gemini)	Soft Voting Explainable AI
Forensic Analytics	BeautifulSoup4, dnspython, Pillow, ImageHash	urllib, Levenshtein, socket, ssl, whois/rdap	SSRF Shield Anti-Spoofing
Cryptographic Ledger	Python `hashlib`, RFC 8785 Canonical JSON	SHA-256 Chaining, PoA Validator Node Engine	Sec 65B Evidence Act
Web Portal (Frontend)	HTML5, Vanilla ES Modules, Modern CSS3	Web Speech API, Web Audio API, UX4G Languages	UX4G 3.0 WCAG 2.1 AAA
Browser Extension	Chrome Manifest V3, Service Worker, Content Scripts	Chrome Tabs/Storage/Action APIs, MutationObserver	MV3 Secure Sandboxed
DevOps & Hosting	Docker Multi-Stage, Render Cloud Web Service, Git	Linux Debian Bookworm, Gunicorn/Uvicorn workers	Zero-Downtime CD

DEFENSE-IN-DEPTH

Detailed Subsystem Specifications

Backend Modules, Forensic Pipeline & Algorithmic Implementation

4. 15-Stage Forensic Analysis Pipeline

1. Canonical URL Normalizer `url_normalizer.py`

Performs recursive percent-decoding, punycode resolution, scheme validation, whitespace scrubbing, and path normalization to expose obfuscated bypass attempts.

2. Homoglyph & Zero-Width Detector `homoglyph_analyzer.py`

Detects Cyrillic, Greek, and Latin confusable Unicode glyphs (e.g., Cyrillic 'a' replacing Latin 'a') alongside invisible zero-width separators (U+200B, U+200C).

3. Typosquatting Engine `typosquat_engine.py`

Evaluates Levenshtein edit distance, bitsquatting, character omissions, repetitions, transpositions, and top-level domain spoofing (e.g., `g0v.in`, `pmkisan.com`).

4. Sovereign Brand Matcher `brand_engine.py`

Indexes 25+ certified sovereign portals (PM-Kisan, IncomeTax, UIDAI, DigiLocker, Parivahan, EPFO) with fuzzy brand injection heuristics against unauthorized hosts.

5. SSRF-Shielded Safe Crawler `safe_crawler.py`

Asynchronous HTTP crawler equipped with IP pre-resolution blocking loopbacks (`127.0.0.1`), RFC 1918 private subnets, and AWS/GCP metadata endpoints (`169.254.169.254`).

6. DOM Sensitive Form Harvester `dom_analyzer.py`

Inspects DOM trees for deceptive credential fields harvesting citizen identities: Aadhaar (12-digit), PAN, Bank Account, Debit/Credit Card, Password, and SMS OTP.

7. Visual Perceptual Hash Engine `visual_analyzer.py`

Renders DOM snapshots and computes 64-bit Perceptual Hash (pHash) and Difference Hash (dHash), flagging layout cloning with Hamming distance < 10 against authentic sites.

8. Redirect Trampoline Unroller `redirect_unroller.py`

Follows HTTP 301/302/307 redirect chains and parses HTML `<meta http-equiv="refresh">` plus inline JS location assignments to uncover concealed landing pages.

9. DNSSEC & MX Mail Security `dns_security_analyzer.py`

Queries authoritative nameservers via `dnspython` for MX records, SPF, DMARC, and DNSSEC. Flags throwaway phishing hosts lacking operational mail infrastructure.

10. RDAP & Sovereign Age `network_analyzer.py`

Extracts domain registration age via ICANN RDAP protocol. Domains registered < 30 days mimicking public brands receive critical risk escalation.

11. Threat Intelligence Hub `reference_database.py`

Queries local high-speed threat caches and OSINT reputation blacklists for known malicious command-and-control (C2) domains and phishing campaigns.

12. Content Similarity Engine `content_similarity.py`

Calculates n-gram Jaccard text similarity against authentic government portal corpora to detect scraping and verbatim identity theft of official disclaimers.

13. OSINT Internet Search Engine `internet_search_engine.py`

Evaluates domain search indexing authority, visibility, and consumer fraud reports across public search telemetry.

14. Fusion Risk Engine `fusion_engine.py`

Synthesizes all 13 module signals using calibrated Bayesian weightings into a normalized 0–100 Risk Score, classifying sites into `LEGITIMATE`, `SUSPICIOUS`, or `PHISHING_CLONE`.

5. AI & Machine Learning Architecture

Sovereign ML Soft Voting Ensemble

- **Architecture**: Soft Voting Ensemble combining **XGBoost (250 estimators)** and **Random Forest (180 estimators)**.
- **Features**: 15 engineered features including Shannon entropy, subdomain depth, lexical token ratio, sensitive input flags, and RDAP age.
- **Explainability**: Computes `top_contributing_factors` translating Gini impurity reductions into natural language citizen warnings.
- **Inference Speed**: Sub-12ms execution cached via serialized Joblib artifacts.

Generative AI Agent & Semantic Synthesis

- **Model Provider**: Google Gemini Pro / Flash via `google-genai` SDK.
- **Functionality**: Analyzes DOM layout, extracted text, and form inputs to explain social engineering tactics used by attackers.
- **Deterministic Fallback**: Built-in deterministic synthesis engine ensuring 100% operational reliability even when cloud AI APIs are offline or unconfigured.

UX4G STANDARD

Frontend, Extension & Legal Compliance

UX4G 3.0 Standard, MV3 Browser Extension & Section 65B Forensic Ledger

6. UX4G 3.0 Multilingual Frontend Portal

The web portal is built strictly adhering to the **UX4G Design System (ux4g.gov.in)** guidelines created by the Ministry of Electronics and Information Technology (MeitY) and National Informatics Centre (NIC).

- **12 Scheduled Indic Languages + English**: Supports Hindi, English, Bengali, Tamil, Telugu, Marathi, Gujarati, Kannada, Malayalam, Punjabi, Odia, and Assamese. All dynamic DOM nodes, badges, advisory text, and forensic layer summaries switch instantly in-memory with zero page reloads.
- **WCAG 2.1 AAA Accessibility Engine**: Features an accessible side drawer (keyboard shortcut `Ctrl+F2`) equipped with 5 color contrast modes (Normal, Monochrome, High Saturation, Low Saturation, Dark Mode, Invert) and 6 typography adjusters (Bigger Text, Line Height, Text Spacing, Highlight Links, OpenDyslexic Font, Hide Images).
- **Natural Acoustic & Voice Synthesis**: Integrates the Web Speech API with Indic BCP-47 locale tags (`hi-IN`, `ta-IN`, etc.) and Web Audio API synthesized acoustic chimes (Harmonic Triad for safe, descending tritone siren for threats).
- **Zero-Framework Performance**: Built using pure ES Modules and lightweight CSS Custom Properties, achieving sub-second first contentful paint (FCP) and a 100/100 Lighthouse performance and accessibility rating.

7. Chrome Browser Extension (Manifest V3)

Component	Responsibilities	Technical Implementation
Background Worker (`background.js`)	Monitors tab updates and activations, maintains in-memory scan cache, drives browser action badges.	Manifest V3 Service Worker, `chrome.tabs.onUpdated`, `chrome.action.setBadgeText`, dual-tier API fetch controller.
Content Script (`content.js`)	Injects non-intrusive warning banners on phishing sites and safe toasts on authentic `.gov.in` portals.	Scoped DOM injection (`#govshield-alert-root`), isolated CSS styles, 12-language mini selector, direct `cybercrime.gov.in` link.
Popup Controller (`popup.js`)	Displays active tab forensic score, collapsible 5-layer details, and voice narration.	Minimalist UX4G popup UI, optimistic client-side heuristics render (<20ms), collapsible dropdown, localized audio playback.
Dual-Tier Fallover	Guarantees zero-failure scanning regardless of network or local development states.	Fast local probe (1000ms) → Cloud Production API (8000ms) → Instant Client-Side Heuristic Fallback.

8. Legal Admissibility & Proof-of-Authority (PoA) Blockchain Ledger

Section 65B Indian Evidence Act Compliance: To allow cybersecurity incident reports to be presented as admissible digital evidence in Indian courts of law, GovShield implements a tamper-proof cryptographic ledger engine (`blockchain_ledger.py`).

- **RFC 8785 Canonical JSON Serialization**: Eliminates cryptographic ambiguity by standardizing key sorting, UTF-8 normalization, and whitespace before hashing.
- **SHA-256 Block Chaining**: Each threat detection generates an immutable block containing the previous block hash, ISO 8601 timestamp, target entity, forensic feature breakdown, raw DOM snapshot SHA-256 digest, and validator identity (`NIC-DELHI-ROOT-01`).
- **CERT-In Standard Dossier**: Exports standardized, pre-formatted incident dossiers ready for automated submission to CERT-In (`incident@cert-in.org.in`) and the National Cyber Crime Reporting Portal (`cybercrime.gov.in`).

9. Production Deployment & Performance Verification

Metric / Environment	Value / Result	Verification Mechanism
Automated Unit Test Suite	**36 / 36 Tests Passed** (12.15s)	`python -m unittest discover backend/tests`
System Scenario Verification	**7 / 7 Scenarios Passed**	`python test_system.py` (Genuine, Typosquat, Homoglyphs, Clones)
Cloud Deployment	Production Ready	Hosted on Render Cloud Web Service with continuous deployment from `main`
Repository	`https://github.com/Code-Xceed/sih-web`	Clean working tree, branch `main`, commit `3938b97`